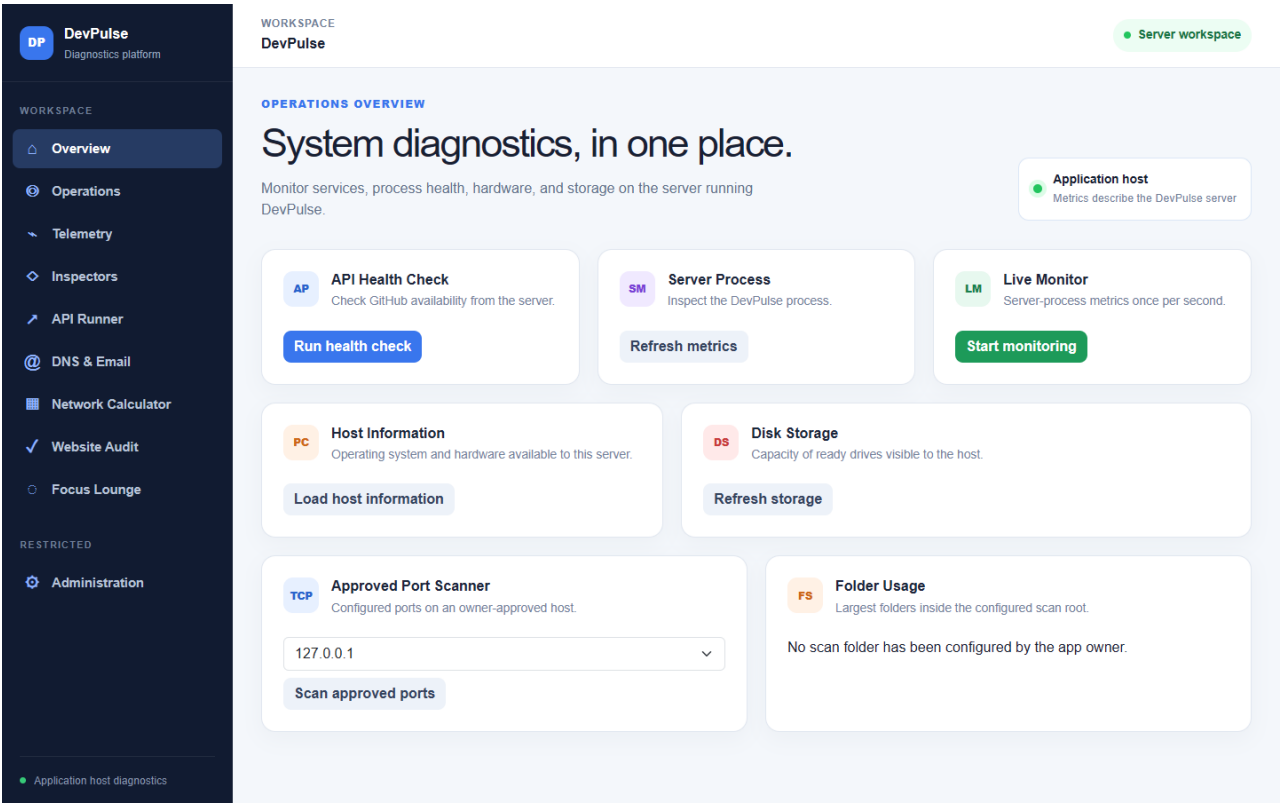


Project Description

A unified workspace for IT diagnostics, website assurance, network planning, API inspection, and developer focus.

Release	Version 1
Platform	.NET 9 Blazor Server
Audience	Technical and nontechnical presentation audiences
Status	Presentation and deployment package



DevPulse Version 1 workspace

CREDITS

Project creators

Add the names of the DevPulse student team and project advisor before sharing or presenting this document.

Role	Name
Student 1	_____
Student 2	_____
Student 3	_____
Advisor	_____

Where to edit the names

Open page 2 in Microsoft Word. Click the blank line in the Name column, delete the underscores, and type the correct name. Repeat this for all three students and the advisor.

Executive summary

DevPulse Version 1 is a .NET 9 Blazor Server application that brings common IT and developer checks into one controlled web workspace.

The problem

Developers and small operations teams often move between operating-system utilities, network calculators, command-line probes, API clients, website-audit tools, credential-sensitive dashboards, and music applications. That switching makes demonstrations harder and evidence more fragmented.

The response

DevPulse combines those workflows without pretending to be an enterprise monitoring replacement. It emphasizes bounded execution, clear ownership, visible safety limits, locally understandable results, and a user interface that can be presented to both technical and nontechnical audiences.

Version 1 value proposition

One web workspace can observe its host, test approved services, inspect local data, plan networks, audit owned websites, exercise APIs, and support focused work.

Release outcome

- A coherent navigation model across diagnostics, assurance, planning, inspection, and focus tools.
- A server-side safety model for outbound requests and administrative actions.
- A Docker deployment contract for Render or another container host.
- Automated regression, build, formatting, and published HTTP smoke evidence.

Users and core workflows

DevPulse is designed for learning, demonstration, small-team operations, and personal development environments.

User	Need	DevPulse workflow
Developer	Check an API or website while coding	API Runner, Website Audit, telemetry, Focus Lounge
IT student	Learn subnets and diagnostic behavior	Network Calculator, ping, DNS, TLS, TCP scans
Small-team operator	See host and service condition	Overview, Operations, telemetry, alerts
Presenter or reviewer	Show evidence and system boundaries	Dashboard, audit results, release status, administration export
Music-enabled user	Stay focused without losing the active Spotify device	Focus Lounge, Spotify Connect, optional browser playback

Typical end-to-end scenario

1. Confirm application and host status on Overview.
2. Calculate the target subnet without manual binary arithmetic.
3. Run approved connectivity checks from the DevPulse host.
4. Audit an owned website and review browser, header, and accessibility evidence.
5. Exercise a public API collection with bounded assertions.
6. Use Focus Lounge while Spotify continues on the selected phone or computer.

Version 1 capability map

The product is organized around five verbs that are easy to explain during a presentation.

Pillar	Modules	Representative outputs
Observe	Overview, Telemetry, Administration	Host metrics, storage, processes, request traces, incidents
Test	Operations, Website Audit, API Runner, DNS & Email	Latency, status, TLS, crawl results, assertions, DNS policy
Inspect	JSON, JWT, SHA-256, encrypted files	Trees, decoded claims, hashes, reversible encrypted packages
Plan	Network Calculator	IPv4/IPv6 ranges, VLSM, routes, bandwidth, MTU and MSS
Focus	Focus Lounge and Spotify	Timers, breathing, ambience, notes, library and playback controls

Notable feature depth

- Website auditing includes crawl limits, headers, JSON checks, optional browser evidence, accessibility, screenshots, visual baselines, and scheduled uptime.
- Spotify includes current playback, active-device following, search, Liked Songs, albums, playlists, queue, shuffle, repeat, transfer, and opt-in browser audio.
- Networking includes both active checks and offline calculations, with explicit restrictions for private and reserved targets.

Architecture and execution model

Blazor Server keeps application logic and credentials on the server while the browser receives an interactive UI over a live circuit.

Layer	Responsibility
Browser	Renders Razor components, captures user actions, maintains local focus notes, and hosts optional Spotify browser audio
Blazor application	Coordinates pages, authentication, validation, diagnostics, and near-real-time UI updates
Application services	Perform system, storage, network, website, DNS, API, Spotify, telemetry, and administration work
Local data directory	Stores encrypted Spotify sessions, data-protection keys, telemetry history, and selected baselines
Approved external services	Public websites, APIs, DNS, Spotify, alert destinations, and optional OTLP collector

Critical interpretation

Every host diagnostic describes the machine running DevPulse. A Render deployment reports its container storage, processes, network path, and reachability. It cannot inspect a visitor's computer without a future authenticated remote agent.

Deployment flow

```
Local source -> .NET 9 publish -> Docker image -> HTTPS hosting service
                                     -> persistent /app/data
```

Technology stack

- ASP.NET Core and Blazor Server on net9.0.
- OpenTelemetry instrumentation with optional OTLP export.
- Node.js and Chromium for optional browser audits.
- Docker multi-stage image running as a non-root user.

Safety and trust model

Diagnostics are useful only when the application prevents them from becoming an unrestricted proxy, scanner, credential sink, or destructive control surface.

Boundary	Version 1 control
Outbound targets	Public-target validation, pinned addresses, exact owner allowlists, redirect blocking
Request volume	Timeout, response-size, request-count, concurrency, port, and crawl caps
Credentials	User Secrets or environment variables; masked transient API authorization values
Spotify sessions	Encrypted server-side tokens, opaque browser cookie, seven-day maximum session
Administration	Separate administrator cookie, antiforgery protection, rate limiting, audit history
Process termination	Disabled by default and requires explicit owner enablement
Production routes	Diagnostic surfaces can return 404 when Diagnostics is disabled

Deployment recommendation

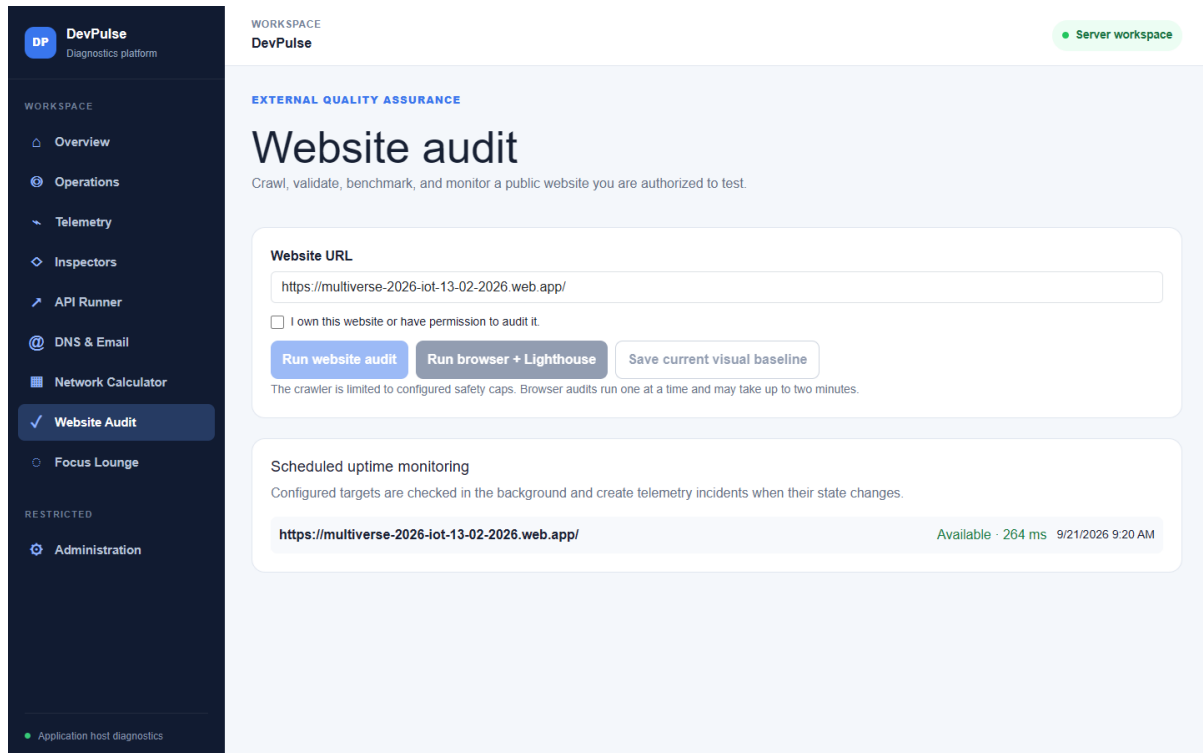
Public deployments should put diagnostic pages behind authenticated access or a private gateway. Spotify authentication is separate and does not grant administrator rights.

Data minimization

- Uploaded inspector files are processed in memory rather than retained as user content.
- API collections exist only in the active Blazor circuit.
- Query strings and authorization data are excluded or redacted from telemetry and exports.

Website and API assurance

DevPulse turns a public site or API that the user owns into a repeatable demonstration target.



Website Audit combines bounded server checks with optional browser evidence.

Website Audit outputs

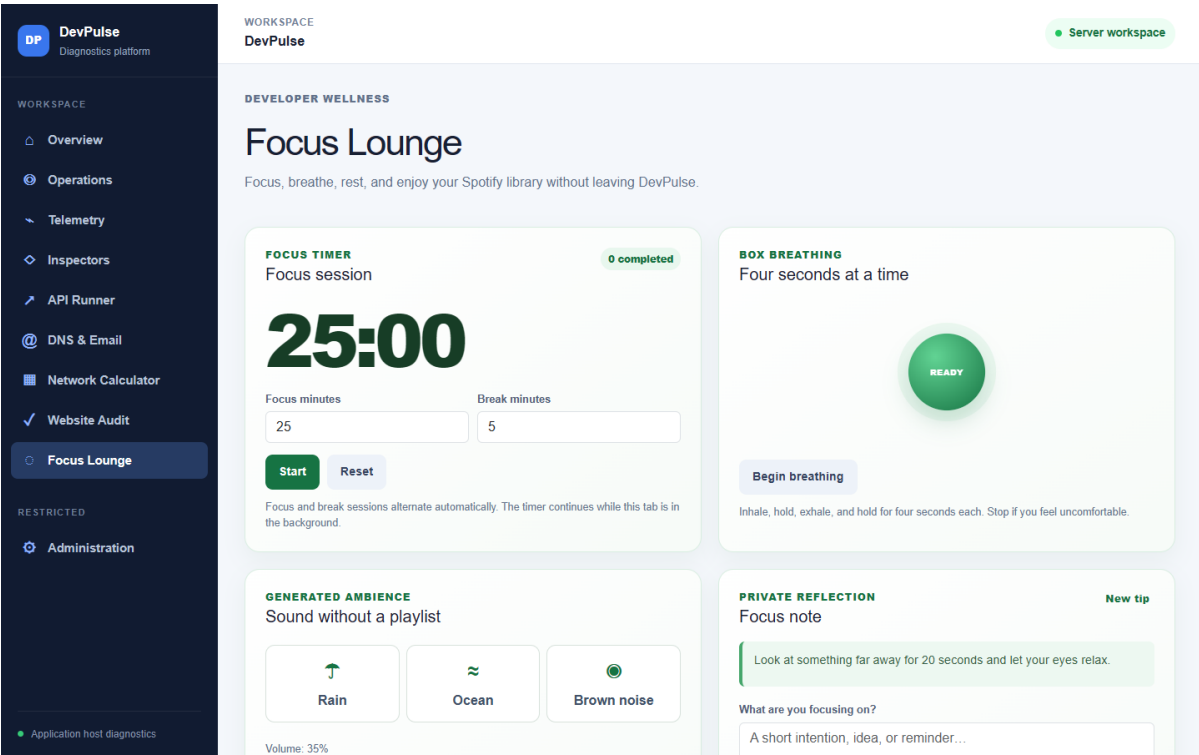
- HTTP status, redirects, timing, headers, metadata, resources, and crawl findings.
- Optional browser run with screenshot, accessibility evidence, performance signals, and visual comparison.
- Scheduled availability checks that create telemetry incidents when state changes.

API and DNS outputs

- Up to ten temporary API requests with methods, headers, bodies, and assertions.
- Public DNS inspection for address, mail, SPF, DMARC, DKIM, and CAA records.
- Strict write-method and private-address boundaries to preserve owner intent.

Developer wellness and Spotify

Focus Lounge keeps restorative tools and music control beside technical work without forcing playback into the browser.



Focus Lounge combines a timer, breathing, generated ambience, reflection, and Spotify.

Experience	Behavior
Focus timer	Alternates focused work and break periods while the tab remains open
Box breathing	Guides four equal phases and can be stopped at any time
Generated ambience	Produces rain, ocean, and brown-noise sound without an external playlist
Private reflection	Keeps the note in browser-local storage
Spotify Connect	Follows the active desktop, phone, speaker, or browser device
Browser playback	Requires Premium and explicit activation by the user

VERIFICATION

Quality and release evidence

Version 1 was validated at service, integration, formatting, publishing, and user-interface levels.

Check	Result	Coverage
Automated regression executable	79 / 79 passed	Spotify parsing and sessions, cryptography, safety rules, audit behavior, telemetry, APIs, DNS
Release build	0 warnings, 0 errors	Complete net9.0 solution build
Formatting	Verified	Repository source formatting
Published HTTP smoke suite	Passed	Routes, static assets, diagnostics gate, OAuth/PKCE, scopes, token endpoint, proxy handling, logout
Visual review	Passed	Desktop navigation and presentation-critical pages

Release statement

The software package is technically ready for a controlled presentation and deployment. Public promotion still depends on securing diagnostic routes and resolving the current Google Safe Browsing or Search Console warning.

Demonstration evidence

- Overview provides immediate system context.
- Network Calculator produces deterministic examples without contacting a target.
- Website Audit demonstrates permission-aware external assurance.
- Focus Lounge provides a clear, nontechnical closing experience.

Known limitations and release notes

These boundaries should be stated during a presentation so that expectations match what Version 1 actually does.

Limitation	Practical meaning
Server-side visibility	Hosted metrics and storage belong to the Render container, not the visitor
Private network reachability	A hosted container cannot normally reach a home or campus LAN
Spotify near-real-time polling	External device changes reconcile periodically because there is no playback webhook
Spotify account constraints	Premium is required for playback controls and browser audio; development apps may restrict users
Database health depth	Version 1 performs TCP acceptance or unauthenticated Redis PING, not SQL query analysis
Single-instance storage	Local encrypted sessions and journals are not shared across replicas
Public reputation review	The current public hostname must clear browser reputation review before broad sharing

Why these limits are useful

They keep Version 1 understandable and safer while establishing clear work for a future remote agent, shared storage, database monitoring, and role-based access.

Presentation route and Version 2

A short live path shows the product clearly without depending on every external integration.

Suggested live demonstration

1. Overview: explain host identity and the difference between local and hosted metrics.
2. Network Calculator: calculate 192.168.10.25/24 and point out the usable range.
3. Website Audit: run or show cached evidence for a website you own.
4. API Runner: perform one public GET request and explain write-method allowlists.
5. Focus Lounge: start the timer, demonstrate breathing, and show Spotify device continuity.

Version 2 candidates

Candidate	Value
SQL Server health and performance monitor	Adds query latency, connection, wait, storage, and backup visibility
IT asset inventory and IP address management	Tracks devices, owners, addresses, subnets, and lifecycle
Authenticated remote monitoring agent	Collects workstation or server metrics beyond the hosted container
Incident response and backup verification	Adds runbooks, evidence, recovery checks, and notification workflows
Role-based access control	Separates viewers, operators, auditors, and administrators

Version 1 conclusion

DevPulse Version 1 establishes the product language, navigation, safety model, deployment contract, and quality baseline needed for these additions.